

How PFAS Became a Top Priority on the Wastewater Agenda



Per- and Polyfluoroalkyl Substances (PFAS), often referred to as ‘forever chemicals’, have emerged as one of the most pressing challenges facing the global wastewater management industry. Although these synthetic compounds have been in use since the 1940s across industrial applications, consumer products, and firefighting foams, their environmental and regulatory implications remained relatively under-recognized for decades. But all that changed in the late 2010s as scientific evidence highlighted their persistence in the environment and potential risks to human health. What was once considered a niche environmental issue has now evolved into a top priority for regulators, utilities, and wastewater operators worldwide.

Why PFAS Moved into the Spotlight

PFAS gained widespread attention as environmental monitoring expanded and these chemicals were increasingly detected in drinking water supplies, groundwater, rivers, lakes, and wastewater streams. This revealed how deeply embedded they had become in the global water cycle. Research raised concerns about PFAS’ ability to persist in the environment, accumulate over time, and potentially impact human health.

At the same time, regulatory scrutiny intensified. In the United States, the U.S. Environmental Protection Agency (EPA) introduced enforceable limits and tightened discharge expectations under federal water laws. In the European Union, the European Chemicals Agency (ECHA) and the European Commission pursued a more precautionary approach, strengthening restrictions and tightening water and wastewater regulations. PFAS is no longer viewed as just another emerging contaminant - it has become a major regulatory, operational, and environmental challenge that wastewater utilities and industrial operators can no longer afford to ignore.

How the Wastewater Industry Is Responding

PFAS presents a challenge for wastewater operators because conventional treatment systems were never designed to remove it. These compounds are highly resistant to natural degradation and are not effectively removed through primary, secondary, or many advanced treatment processes. As a result, treatment plants designed for bulk pollutant removal are now being asked to manage contaminants at extremely low concentrations.

This has accelerated the need for the wastewater industry to explore new treatment approaches. Wastewater operators are deploying adsorption-based technologies such as granular activated carbon (GAC) and ion exchange resins to capture PFAS. Alongside these established systems, emerging technologies, including plasma treatment, electrochemical oxidation, and thermal destruction are gaining traction as potential long-term options capable of breaking down PFAS rather than simply removing them. While many of these technologies are still in early-stage commercialization, they are attracting growing interest as the industry searches for more effective ways to manage PFAS contamination.

It is becoming increasingly clear that PFAS cannot be tackled through wastewater treatment alone. Wastewater operators are working more closely with industrial users to reduce PFAS at the source through pre-treatment programs, stricter discharge controls, and improved monitoring. This shift reflects a broader industry view that effective PFAS management requires action across the entire wastewater value chain, from industrial production and discharge to treatment and final disposal.

Digital technologies such as advanced monitoring tools, data analytics, and improved sampling techniques are helping wastewater operators to gain greater visibility into PFAS contamination, strengthen compliance efforts, and improve operational decision-making. Detecting PFAS at extremely low concentrations remains technically challenging and expensive, but digital tools are helping organizations respond more effectively.

Beyond Treatment: The Growing Biosolids Challenge

Recent studies and regulatory reviews have shown that PFAS is not just a water treatment issue. These chemicals are increasingly being detected in wastewater streams, sludge, and biosolids, extending the challenge beyond water treatment alone. This has led to tighter restrictions in several regions on biosolids disposal and land application, adding another layer of complexity to wastewater management.

Conclusion

PFAS has evolved from a little-known contaminant into one of the defining challenges for the wastewater industry. It is driving investment in new treatment technologies, strengthening regulatory frameworks, and increasing focus on source control. PFAS reflects a broader shift in the industry, from managing conventional pollutants to addressing persistent contaminants that require new technologies, new operating models, and deeper coordination across the value chain. How the industry responds to PFAS will shape the future of wastewater management.